

MATH 2130 - Week 4

Eigenvalues and Eigenvectors

1 Overview

The “second half” of matrix theory is concerned with finding solutions to the eigen-equation:

$$Av = \lambda v$$

It’s hard to overstate just how important this equation is. This week will just be a start of an incredibly deep and powerful group of ideas, all connected to the **eigenvalues** λ and **eigenvectors** v of the previous equation. This is unquestionable the most important material of the course.

We will focus primarily on “small” matrices and focus on understanding how to explicitly compute eigenvalues and eigenvectors in the 2×2 and 3×3 case. We will see how to use Sage to work with larger matrices. Once we have become familiar with the basics, we will talk about **eigenspaces** and see their connection to many of the topics covered earlier in the course. Since **real-valued matrices** can have complex eigenvalues, we will need to introduce **complex numbers**.

One of the most important applications of eigenvalues and eigenvectors is **diagonalization** of a square matrix. We will learn when this is possible and how to do it. This leads naturally to the idea of **similarity**. Building off of this, we will talk about **change of basis** and how to explicitly compute the matrix with respect to a different set of basis vectors.

We will take a bit of a detour into applications and look at how eigenvalues and eigenvectors allow us to solve **difference equations** and **differential equations**. We will focus primarily on concrete examples and not spend too much time developing the more theoretical side of things.

Finally, we will consider **complex-valued matrices**. We will discuss what happens in the complex case, focusing primarily on **Hermitian** and **Unitary** matrices. Again, we will focus on concrete examples and not spend too much time on the more theoretical side.

I will not be in town on Friday, but we **WILL** still have class. You will take this time to work together on the Take-Home Exam, which is due next Wednesday.

2 Schedule and Checklist

Here is a brief schedule for the week. Please note that this is subject to change, so you need to adjust to what we do in class, regardless of what it says here. Think of this as a pretty accurate guideline, though.

- (M) Eigenvalues and Eigenvectors.
- (T) Diagonalization and $A = SAS^{-1}$ Factorization.

- (W) Algebraic and Geometric Multiplicity.
- (R) Similar Matrices.
- (F) Work on Take-Home Exam.

Here is a checklist of things to do this week. This is meant to help you keep track of what you've completed and what you still need to work on.

- ☐ Read these notes and ask any questions that you have about anything.
- ☐ Read the sections from Austin and Olver.
- ☐ Complete all of the homework exercises.

3 Key Concepts

- Eigenvalues
- Characteristic Polynomial
- Eigenvectors
- Eigenspaces
- Diagonalization
- Similar Matrices
- Change of Basis
- Difference and Differential Equations
- Complex-Valued Matrices
- Hermitian and Unitary Matrices

4 Important Definitions, Formulas, and Theorems

- The number λ is an **eigenvalue** of the $n \times n$ matrix A if and only if

$$\det(A - \lambda I) = 0$$

This is called the **characteristic equation** for the matrix A . The left-hand side is a polynomial in λ , called the **characteristic polynomial**, denoted by $p(\lambda)$. This means that the eigenvalues of A are the roots of $p(\lambda)$.

- Given an eigenvalue λ of the matrix A , an associated **eigenvector** is a non-zero vector v such that

$$Av = \lambda v$$

Geometrically, v is an eigenvector of A if multiplying v by A doesn't change the direction of v (though it might reverse the direction if λ is negative). So eigenvectors stay pointing in their same direction after being multiplied by A . They are just scaled by the eigenvalue, but not moved.

Connected to the previous comment, note that if v is an eigenvector, then so is kv for any non-zero scalar k . This follows easily from the definitions of eigenvalues and eigenvectors and properties of matrix multiplication:

$$A(kv) = k(Av) = k(\lambda v) = \lambda(kv)$$

This gives us some flexibility in choosing the components of v and we will usually choose them in such a way that they are all relatively small integers.

We will find the eigenvectors by solving the system of equations

$$(A - \lambda I)v = \vec{0}$$

- The sum of the eigenvalues of a matrix A is equal to the sum of the diagonal entries of A :

$$\lambda_1 + \lambda_2 + \cdots + \lambda_n = a_{11} + a_{22} + \cdots + a_{nn}$$

The sum on the right-hand side is called the **trace** of the matrix A :

$$\text{tr}(A) = \sum_{i=1}^n a_{ii}$$

- The product of the eigenvalues of a matrix A is equal to the determinant of A :

$$\prod_{i=1}^n \lambda_i = \det(A)$$

An immediate consequence of this is that if any of the eigenvalues of A is equal to 0, then $\det(A) = 0$ and A is a singular matrix. Similarly, if none of the eigenvalues are equal to 0, then $\det(A) \neq 0$ and A is nonsingular.

- If the matrix A is triangular (so it can be upper-triangular, lower-triangular, or diagonal), then the eigenvalues of A are the same as the diagonal entries of A .
- If λ is an eigenvalue of the $n \times n$ matrix A , with eigenvector v , then λ^2 is an eigenvalue of the matrix A^2 , and the corresponding eigenvector is still v . This follows easily from the definitions of eigenvalues and eigenvectors and properties of matrix multiplication:

$$A^2v = A(Av) = A(\lambda v) = \lambda(Av) = \lambda(\lambda v) = \lambda^2v$$

This process can be easily extended to show that λ^k is an eigenvalue of A^k for any positive integer power k .

Geometrically, all this says is that if multiplication by A leaves the direction of v unchanged, then multiplying by A again (and again,...) will continue to leave the direction unchanged.

- Suppose the $n \times n$ matrix A has n linearly independent eigenvectors. Form the matrix S having these n eigenvectors as columns. Then S is invertible and

$$S^{-1}AS = \Lambda = \begin{bmatrix} \lambda_1 & & & & \\ & \lambda_2 & & & \\ & & \lambda_3 & & \\ & & & \ddots & \\ & & & & \lambda_{n-1} & \\ & & & & & \lambda_n \end{bmatrix}$$

The matrix Λ is a diagonal matrix with the eigenvalues of A along the diagonal. When this happens, we say that the matrix A is **diagonalizable** and call the matrix Λ the **diagonalization** of A .

The above equation $S^{-1}AS = \Lambda$ can be rewritten in either of the following equivalent forms:

$$AS = S\Lambda \quad \text{or} \quad A = S\Lambda S^{-1}$$

- If the $n \times n$ matrix A has n distinct eigenvalues, then the n eigenvectors are automatically linearly independent, and so A is diagonalizable.
- If λ_i is a repeated root of the characteristic polynomial $p(\lambda)$, then $p(\lambda)$ can be written as $p(\lambda) = (\lambda - \lambda_i)^k q(\lambda)$ where $k > 1$ and $q(\lambda)$ is a polynomial of degree $n - k$. The number k is called the **algebraic multiplicity** of the eigenvalue λ_i .
- The nullspace of $A - \lambda I$ is called the **eigenspace** of the eigenvalue λ and is denoted E_λ . The dimension of E_λ is called the **geometric multiplicity** of the eigenvalue λ .
- The matrix A is diagonalizable if and only if the algebraic multiplicity of every eigenvalue equals the geometric multiplicity.

5 Examples

- We'll work with the following 2×2 matrix

$$A = \begin{bmatrix} 4 & -5 \\ 2 & -3 \end{bmatrix}$$

It has characteristic polynomial

$$p(\lambda) = \lambda^2 - \lambda - 2$$

which leads to the eigenvalues

$$\lambda_1 = -1, \quad \text{and} \quad \lambda_2 = 2$$

The associated eigenvectors are

$$v_1 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{and} \quad v_2 = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$

- We'll also look at the following 3×3 symmetric matrix

$$B = \begin{bmatrix} 1 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

It has characteristic polynomial

$$p(\lambda) = -\lambda^3 + 4\lambda^2 - 3\lambda$$

which leads to the eigenvalues

$$\lambda_1 = 0, \quad \lambda_2 = 1, \quad \text{and} \quad \lambda_3 = 3$$

The associated eigenvectors are

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad v_2 = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}, \quad \text{and} \quad v_3 = \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix}$$

- The following 2×2 matrix is **not** diagonalizable:

$$C = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$$

It has characteristic polynomial $p(\lambda) = \lambda^2$ so $\lambda = 0$ is a repeated eigenvalue (the **algebraic multiplicity** of the eigenvalue is 2) but it only has one linearly independent eigenvector (the **geometric multiplicity** of the eigenvalue is 1):

$$v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

This means that we do not have enough eigenvectors to construct the matrix S and so the matrix A is not diagonalizable.

6 Sage Code

6.1 Commands

These are the most common Sage commands that we will use this week. Think of this more as a reference. It is usually easier to understand what the commands do by seeing them used in example code, which is given below.

- `.charpoly()`

This calculates the characteristic polynomial of the matrix it is appended to. By default, the polynomial is in the variable x (instead of λ).

- `.fcp()`

This returns the factored characteristic polynomial of the matrix it is appended to. Again, the default variable is x .

- `.eigenvalues()`

This returns a list of all distinct eigenvalues of the matrix that it is appended to.

- `.trace()`

This returns the trace of the matrix it is appended to.

- `.eigenvectors_right()`

This returns a list of lists. Each sublist contains three pieces of information: a (distinct) eigenvalue, a basis for the corresponding eigenspace, and the algebraic multiplicity of the eigenvalue. Each element in the basis is an eigenvector for the given eigenvalue. If the algebraic multiplicity is equal to the number of vectors in the basis of the eigenspace, for each eigenvalue, then the matrix is diagonalizable.

- `.eigenspaces_right()`

This gives essentially the same information as the `.eigenvectors_right()` command, just in a way that is a little easier to read. You'll probably just work with the `.eigenvectors_right()` command.

- `.eigenmatrix_right()`

This returns a list with two matrices, Λ and S , corresponding to the matrix it is appended to. If the original matrix is A , then `A.eigenmatrix_right()` returns a list $[\Lambda, S]$ such that $\Lambda = S^{-1}AS$. Please note the order.

6.2 Example Code

See the folder in the Google Drive for examples of Sage code corresponding to this material.

7 Additional Comments

- Eigenvalues and eigenvectors are the **MOST** important concept covered in this course. They are absolutely crucial and will show up over and over again in future math courses and in applications.
- The only way to appreciate the importance of these ideas is to spend some time with them. It will take a little while to gain an appreciation for them, but it is well worth the effort.
- At heart, eigenvalues and eigenvectors “show” you what a matrix is really doing (thinking of a matrix as representing a linear transformation from a vector space to itself). The German word “eigen” roughly translates to “characteristic,” and in a very real sense the eigenvalues and the eigenvectors give you the characteristic properties of the matrix.
- On top of all of this, they are just incredible objects that simplify and unify much of Linear Algebra and everything that uses it.
- Please keep in mind that while the eigenvalues are fixed (by the entries of the matrix itself) the eigenvalues are determined by a system of equations that has infinitely many solutions. This means that any eigenvalue will have infinitely many eigenvectors, but they will all be nonzero scalar multiples of each other. You'll want to remember this if you're using Sage (or anything else) to check your work because you might have a different eigenvector than what you see elsewhere. But as long as they are scalar multiples of each other, both are correct.

8 Reading Notes

- Olver 8.1 - Linear Dynamical Systems
- Olver 8.2 - Eigenvalues and Eigenvectors
- Olver 8.3 - Eigenvector Bases
- Olver 8.4 - Invariant Subspaces
- Olver 8.5 - Eigenvalues of Symmetric Matrices

9 Homework Exercises

See the “Problem Set 4” file that has been uploaded into the Google Drive.